Ayan Bhardwaj

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EDUCATION

The University of Texas at Austin **Austin, TX**
B.S. Computer Science Honors & B.B.A. Canfield Business Honors (CSB Honors Program) *May 2028*
- GPA: **4.0/4.0** | Coursework: Data Structures (Honors), Discrete Math (Honors), Probability & Statistics, Machine Learning, Calculus III

TECHNICAL SKILLS

Languages: Python, TypeScript/JavaScript, Java, C++ (incl. embedded), SQL, Rust, MATLAB
AI & LLM Workflows: LLM APIs (OpenAI, Anthropic), RAG, agents and tool calling, PyTorch, scikit-learn, OpenCV, data pipelines
Backend, Data & Cloud: REST APIs, FastAPI, Node.js, React, Next.js, PostgreSQL, BigQuery, Docker, AWS, Modal, Vercel
Dev Tools: Git/GitHub, GitHub Actions (CI/CD), automated testing, code review, Claude Code and agentic AI coding tools

EXPERIENCE

Moss (YC F25) **Remote / San Francisco, CA**
Software Engineering Intern *May 2026 – Aug 2026*
- Owned **10** open-source apps end to end, from concept to launch, in TypeScript and Rust on Moss's WebAssembly semantic search SDK (mossJARVIS, Mycelium, Sentry, Thread, StudyOverlay), delivering sub-**10ms** on-device retrieval.
- Tested semantic search accuracy across all **10** builds; the data was used to improve the core SDK's retrieval consistency by **20%**.
- Designed the OpenRouter LLM workflow layer reused across all **10** builds and adopted by **5** fellow engineering interns.
- Drove developer adoption with technical content reaching **5,000+** views, bringing **100+** GitHub stars to the core SDK repo.

Koyal AI (YC F25) **Remote**
AI Engineering and Business Development Intern *May 2025 – Sept 2025*
- Built OpenCV scene classification across **300+** video scenes, cutting manual review time by **70%**, and integrated an emotion classifier mapping 48 emotions to 6 core categories (**95%** lip-sync accuracy).
- Built a production face-swap pipeline with retries and multi-tier fallbacks across **1,000+** images (**90%** accuracy gain).
- Engineered a multi-provider lip-sync pipeline and a GPU load balancer for **12,000+** monthly video requests (**65%** fewer bottlenecks).
- Worked directly with the founders on client management and industry outreach, growing the product waitlist to **500+** users.

MIT Lincoln Laboratory **Cambridge, MA**
Engineering Researcher *Jun 2025 – Aug 2025*
- Owned OncoSense end to end, from optical sensing research to a shipped wearable near-infrared skin monitor integrating custom hardware (SAMD21 MCU, **10**-channel spectral sensor, **850**/**660** nm LEDs over I^2C) with full-stack software.
- Root-caused cross-session calibration drift and rewrote the C++ firmware with ambient-light rejection, auto-gain saturation detection, per-user two-point calibration, and real-time signal quality flags.
- Shipped **3** sensing modes (hydration, surface change, **10**-channel reflectance spectra) to a live Vercel web app, with GitHub Actions CI running unit tests and a firmware-in-the-loop hardware simulator.

PROJECTS

YouBank | Python, TypeScript, Node.js, PostgreSQL, OpenAI/Anthropic APIs, Modal, BigQuery **github.com/AyanBhardwaj1/YouBank**
- Built LLM agent workflows for finance analysts on the OpenAI and Anthropic APIs, with RAG over company filings using hybrid retrieval and cross-encoder reranking (recall@5 from **0.85** to **0.97**).
- Integrated SEC EDGAR, Treasury, BLS, and market data feeds into one PostgreSQL-backed platform with **322** REST API routes.
- Deployed Python ML microservices on Modal behind an authenticated async API; integrated BigQuery for time-series forecasting.
- Set up GitHub Actions CI/CD for cross-platform desktop builds, backed by **1,000+** unit tests and end-to-end suites.

Walk-Free | React, TypeScript, Three.js, Express, PostgreSQL (Neon), Vercel **github.com/AyanBhardwaj1/Walk-Free**
- Built a heat-aware walking route planner that pathfinds over OpenStreetMap pedestrian graphs with modeled building and tree-canopy shade, offering **6** route types including shade-optimized and step-free.
- Shipped live across **7** areas (**6** campuses + Midtown Manhattan) with 3D scenes, a tool-calling LLM assistant, and offline PWA caching.

LEADERSHIP & AWARDS

Advanced Equities Organization **Dallas, TX**
Founder *Aug 2024 – Mar 2026*
- Founded a 501(c) financial literacy nonprofit and grew it to **200+** members; managed a **$15,000** live investment portfolio and ran a stock pitch competition where **300+** interns submitted **100+** reports.

NVIDIA × YTU Startup House AI Engineer Bootcamp 2026: Selected developer from **10,000+** global applicants into an **880**-person cohort across **27** countries; NVIDIA-taught training in LLM internals, production RAG, MCP tool calling, agent loops, LLM evals, and NIM deployment
Y Combinator Startup School 2026: Technical builder cohort, selected from **30,000+** applicants ($25,000 in API credits)
Hackathons: Spurs² Hackathon @ UT Austin (**Top 3** Placement); Meta AI Global Developer Hackathon (Accepted)
Federal Reserve Bank of New York: Fed Challenge Winner, Economics Publication
Organizations: Texas Accelerate, Directed Reading Program (DiRP), CS and Business Association (CSBA), Honors Business Association (HBA)
Interests: NBA (Dallas Mavericks), soccer, pickleball, jersey collecting | Languages: English, Hindi (fluent)